🧠 CORE CS (THEORY)

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Instagram: code.abhii07 (SYNTAX ERROR)

❖Top 50 Operating System Interview Questions

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1. What is an Operating System?

An Operating System (OS) is system software that manages computer hardware and software resources. It acts as an interface between user and hardware. The OS handles processes, memory, files, and devices. Examples include Windows, Linux, and macOS.

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2. What are the main functions of an OS?

The OS manages processes, memory, files, and input/output devices. It provides security and user interface. It ensures efficient resource utilization. It also handles error detection and recovery.

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3. What is a Process?

A process is a program in execution. It includes program code, data, and execution state. Each process has its own resources. OS schedules processes for execution.

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4. What is a Thread?

A thread is a lightweight unit of a process. Multiple threads can exist within a process. Threads share resources of the process. They improve performance and responsiveness.

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5. Difference between Process and Thread

A process has its own memory space. A thread shares memory with other threads. Process creation is heavy, thread creation is light. Threads are faster to switch.

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6. What is Multitasking?

Multitasking allows multiple programs to run simultaneously. The CPU switches between tasks quickly. It improves system efficiency. Common in modern operating systems.

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7. What is Multiprocessing?

Multiprocessing uses multiple CPUs or cores. Tasks are executed in parallel. It increases system performance. Used in servers and high-end systems.

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8. What is Multithreading?

Multithreading allows multiple threads in a process. Threads run concurrently. It improves CPU utilization. Common in applications like browsers.

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9. What is Scheduling?

Scheduling decides which process executes next. It is managed by the OS scheduler. It improves CPU efficiency. Different algorithms are used.

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10. What is a Scheduling Algorithm?

Scheduling algorithms determine process execution order. Examples include FCFS, SJF, and Round Robin. They affect waiting time and turnaround time. Used for fair CPU allocation.

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11. What is FCFS Scheduling?

First Come First Serve executes processes in arrival order. It is simple to implement. It may cause long waiting times. Not efficient for all cases.

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12. What is SJF Scheduling?

Shortest Job First selects the shortest process. It reduces average waiting time. Can be preemptive or non-preemptive. Difficult to predict burst time.

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13. What is Round Robin Scheduling?

Round Robin assigns time slices to processes. Each process gets equal CPU time. Prevents starvation. Common in time-sharing systems.

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14. What is Priority Scheduling?

Processes are scheduled based on priority. Higher priority executes first. Can be preemptive or non-preemptive. May cause starvation.

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15. What is Starvation?

Starvation occurs when a process never gets CPU time. Low-priority processes suffer. It affects fairness. Aging technique solves it.

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16. What is Aging?

Aging gradually increases priority of waiting processes. It prevents starvation. Ensures fairness. Used in priority scheduling.

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17. What is Deadlock?

Deadlock occurs when processes wait for each other. Resources are never released. System halts execution. Common in multitasking systems.

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18. Conditions for Deadlock

Deadlock requires mutual exclusion, hold and wait, no preemption, and circular wait. All four must occur simultaneously. If one is avoided, deadlock can be prevented.

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19. What is Deadlock Prevention?

Deadlock prevention avoids one of the four conditions. It restricts resource allocation. Ensures system safety. May reduce efficiency.

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20. What is Deadlock Avoidance?

Deadlock avoidance predicts deadlocks before they occur. Uses algorithms like Banker’s Algorithm. Requires prior resource information. More flexible than prevention.

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21. What is Banker’s Algorithm?

Banker’s Algorithm avoids deadlock. It checks safe state before allocation. Used in resource management. Requires maximum resource need.

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22. What is Memory Management?

Memory management handles allocation and deallocation. It ensures efficient memory usage. Prevents memory leaks. Managed by OS.

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23. What is Paging?

Paging divides memory into fixed-size pages. Eliminates external fragmentation. Uses page tables. Improves memory utilization.

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24. What is Segmentation?

Segmentation divides memory into logical segments. Each segment has variable size. Improves program structure. May cause fragmentation.

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25. Difference between Paging and Segmentation

Paging uses fixed-size blocks. Segmentation uses variable-size blocks. Paging avoids external fragmentation. Segmentation reflects program logic.

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26. What is Virtual Memory?

Virtual memory uses disk as extra RAM. Allows large programs to run. Improves memory utilization. Implemented using paging.

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27. What is Thrashing?

Thrashing occurs when system spends more time swapping pages. CPU utilization drops. Happens due to low memory. Affects performance.

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28. What is Page Fault?

A page fault occurs when page is not in memory. OS loads page from disk. Common in virtual memory. Increases execution time.

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29. What is Fragmentation?

Fragmentation is wasted memory space. Internal fragmentation occurs within blocks. External fragmentation occurs between blocks. Reduces efficiency.

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30. What is File System?

File system organizes data storage. It manages files and directories. Provides access methods. Examples include NTFS and FAT.

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31. What is a Directory?

A directory stores file information. Helps organize files. Contains file names and locations. Improves accessibility.

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32. What is I/O Management?

I/O management handles input/output devices. OS uses drivers. Improves efficiency. Manages buffering and caching.

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33. What is Buffering?

Buffering stores data temporarily. It smooths data transfer. Improves speed. Used in I/O operations.

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34. What is Spooling?

Spooling stores data on disk temporarily. Used for slow devices like printers. Improves CPU utilization. Allows multitasking.

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35. What is Context Switching?

Context switching saves process state. CPU switches to another process. Enables multitasking. Adds overhead.

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36. What is Kernel?

Kernel is the core of OS. It controls hardware and system calls. Runs in privileged mode. Essential for OS functioning.

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37. Types of Kernel

Types include monolithic, microkernel, and hybrid. Monolithic is fast but complex. Microkernel is modular. Hybrid combines both.

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38. What is System Call?

System call allows user programs to access OS services. Examples include file and process operations. Acts as interface. Runs in kernel mode.

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39. What is User Mode and Kernel Mode?

User mode has limited access. Kernel mode has full access. OS switches modes for safety. Prevents misuse of resources.

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40. What is Semaphore?

Semaphore is a synchronization tool. Controls access to shared resources. Prevents race conditions. Used in multithreading.

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41. What is Mutex?

Mutex ensures mutual exclusion. Only one thread accesses resource. Prevents conflicts. Used in critical sections.

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42. What is Synchronization?

Synchronization coordinates processes. Prevents data inconsistency. Uses tools like semaphore. Essential in concurrency.

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43. What is Race Condition?

Race condition occurs when processes access shared data simultaneously. Results are unpredictable. Occurs due to lack of synchronization. Needs proper control.

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44. What is Critical Section?

Critical section is code accessing shared resources. Only one process allowed. Prevents race conditions. Uses locks or mutex.

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45. What is Booting?

Booting starts the computer system. Loads OS into memory. Can be cold or warm boot. Initializes hardware.

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46. What is Interrupt?

Interrupt signals CPU to stop current task. Handles urgent events. Improves responsiveness. Can be hardware or software.

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47. What is Polling?

Polling checks device status repeatedly. CPU actively waits. Simple but inefficient. Alternative to interrupts.

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48. What is Time Sharing System?

Time sharing allows multiple users. CPU time is divided. Improves responsiveness. Used in multi-user OS.

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49. What is Distributed Operating System?

Distributed OS manages multiple systems. Appears as single system. Improves reliability. Used in networks.

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50. Why is OS Important?

OS manages resources efficiently. Provides user interface. Ensures security and stability. Essential for system operation.

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